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Non-localized spin valve multi-free-layer reader hybridized with spin orbit torque layers

Abstract

The present disclosure generally relates to a magnetic recording head comprising a read head. The read head comprises a first sensor disposed at a media facing surface (MFS) comprising at least one free layer, a second sensor disposed at the MFS comprising at least one free layer, a first spin generator spaced from the first sensor and recessed from the MFS, and a second spin generator spaced from the second sensor and recessed from the MFS. The first and second spin generators each individually comprises at least one spin orbit torque (SOT) layer. The SOT layer may comprise BiSb. The first and second sensors are configured to detect a read signal using a first voltage lead and a second voltage lead. The first and second spin generators are configured to inject spin current through non-magnetic layers to the first and second sensors using a plurality of current leads.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims benefit of U.S. provisional patent application Ser. No. 63/523,308, filed Jun. 26, 2023, which is herein incorporated by reference.

BACKGROUND OF THE DISCLOSURE
Field of the Disclosure
(1) Embodiments of the present disclosure generally relate to a magnetic recording head comprising a read head, such as a magnetic media drive or magnetic disk drive.
Description of the Related Art
(2) At the heart of a computer is a magnetic disk drive. Information is written to and read from a disk as the disk rotates past read and write heads that are positioned very close to the magnetic surface of the disk. There have been various read head designs proposed to boost linear density based on different physical mechanisms such as a tunneling magneto-resistive (TMR) effect, a magneto-resistance (GMR) effect, an extraordinary magneto-Resistive (EMR) effect, a spin torque oscillator (STO) effect, and non-localize spin valve (NLSV).
(3) In NLSV design, the read head may include two spatially separated tunnel junction structures, with one tunnel junction structure being recessed from a media facing surface (MFS) and the other structure being disposed at the MFS. The recessed structure often comprises a pinned layer and a ferromagnetic layer having a fixed magnetization for spin injection. However, the reader signal output in such read heads is limited by the spin injection efficiency from the pinned layer, which has a spin polarization of less than 1. As such, the relatively low spin current injection results in poor signal output, negatively impacting the read head's ability to accurately read data.
(4) Therefore, there is a need in the art for improved read heads having high signal outputs for use in magnetic recording devices.
SUMMARY OF THE DISCLOSURE
(5) The present disclosure generally relates to a magnetic recording head comprising a read head. The read head comprises a first sensor disposed at a media facing surface (MFS) comprising at least one free layer, a second sensor disposed at the MFS comprising at least one free layer, a first spin orbit torque (SOT) spin generator spaced from the first sensor and recessed from the MFS, and a second spin generator spaced from the second sensor and recessed from the MFS. The first and second spin generators each individually comprises at least one SOT layer. The SOT layer may comprise BiSb. The first and second sensors are configured to detect a read signal using a first voltage lead and a second voltage lead. The first and second spin generators are configured to inject spin current through non-magnetic layers to the first and second sensors using a plurality of current leads.
(6) In one embodiment, a read head comprises a first shield; a second shield, a first non-magnetic layer disposed between the first shield and the second shield, a first sensor disposed between the

first non-magnetic layer and the second shield at a media facing surface (MFS), the first sensor comprising a first free layer, a first spin generator recessed from the MFS, the first spin generator being spaced from the first sensor, wherein the first spin generator comprises a first spin orbit torque (SOT) layer, a third shield disposed over the second shield, a second non-magnetic layer disposed between the third shield and the fourth shield, a second sensor disposed between the second non-magnetic layer and the fourth shield at the MFS, the second sensor comprising a second free layer, and a second spin generator recessed from the MFS, the second spin generator being spaced from the second sensor, wherein the second spin generator comprises a second SOT layer.

(7) In another embodiment, a magnetic recording head comprises a read head, the read head comprising: a first shield, the first shield comprising a first shield notch, a second shield disposed over the first shield, a first non-magnetic layer disposed between the first shield notch and the second shield, a first sensor disposed between the first non-magnetic layer and the second shield at a media facing surface (MFS), the first sensor comprising a first free layer, a first spin generator disposed between the first shield and the second shield recessed from the MFS, the first spin generator being spaced from the first sensor, wherein the first spin generator comprises: a first spin orbit torque (SOT) layer disposed between the first shield and the first non-magnetic layer, and a second SOT layer disposed between the first non-magnetic layer and the second shield, a third shield disposed over the second shield, the third shield comprising a second shield notch, a fourth shield disposed over the third shield, a second non-magnetic layer disposed between the second shield notch and the fourth shield, a second sensor disposed between the second non-magnetic layer and the fourth shield at the MFS, the second sensor comprising a second free layer, and a second spin generator disposed between the third shield and the fourth shield recessed from the MFS, the second spin generator being spaced from the second sensor, wherein the second spin generator comprises: a third SOT layer disposed between the third shield and the second non-magnetic layer, and a fourth SOT layer disposed between the second non-magnetic layer and the fourth shield.

(8) In yet another embodiment, a magnetic recording head comprises a read head, the read head comprising: a first shield, a second shield, a first non-magnetic layer disposed between the first shield and the second shield, a first sensor disposed between the first shield and the second shield at a media facing surface (MFS), the first sensor comprising a first free layer and a second free layer, a first spin generator disposed between the first shield and the second shield recessed from the MFS, the first spin generator being spaced from the first sensor, wherein the first spin generator comprises: a first spin orbit torque (SOT) layer disposed between the first shield and the first non-magnetic layer, and a second SOT layer disposed between the first non-magnetic layer and the second shield, a third shield disposed over the second shield, a second non-magnetic layer disposed between the second shield and the third shield, a second sensor disposed between the second shield and the third shield at the MFS, the second sensor comprising a third free layer and a fourth free layer, and a second spin generator disposed between the second shield and the third shield recessed from the MFS, the second spin generator being spaced from the second sensor, wherein the second spin generator comprises: a third SOT layer disposed between the second shield and the second non-magnetic layer, and a fourth SOT layer disposed between the second non-magnetic layer and the third shield.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) So that the manner in which the above recited features of the present disclosure can be understood in detail, a more particular description of the disclosure, briefly summarized above,

may be had by reference to embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only typical embodiments of this disclosure and are therefore not to be considered limiting of its scope, for the disclosure may admit to other equally effective embodiments.
(2) FIG. **1** illustrates a disk drive embodying this disclosure.
(3) FIG. **2** is a fragmented, cross-sectional side view through the center of a read/write head facing a magnetic media, according to one embodiment.
(4) FIG. **3** illustrates a conventional read head based on NLSV.
(5) FIGS. **4**A-**4**B illustrate a read head, according to one embodiment.
(6) FIGS. **4**C-**4**D illustrate a read head, according to another embodiment.
(7) FIGS. **5**A-**5**B illustrate a read head of a magnetic recording head, according to one embodiment.
(8) FIGS. **5**C-**5**D illustrate a read head of a magnetic recording head, according to another embodiment.
(9) FIGS. **6**A-**6**B illustrate a read head of a magnetic recording head, according to yet another embodiment.
(10) FIGS. **6**C-**6**D illustrate a read head of a magnetic recording head, according to one embodiment.
(11) FIGS. **7**A-**7**B illustrate a read head of a magnetic recording head, according to yet another embodiment.
(12) FIGS. **8**A-**8**B illustrate a read head of a magnetic recording head, according to another embodiment.
(13) To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the figures. It is contemplated that elements disclosed in one embodiment may be beneficially utilized on other embodiments without specific recitation.
DETAILED DESCRIPTION
(14) In the following, reference is made to embodiments of the disclosure. However, it should be understood that the disclosure is not limited to specific described embodiments. Instead, any combination of the following features and elements, whether related to different embodiments or not, is contemplated to implement and practice the disclosure. Furthermore, although embodiments of the disclosure may achieve advantages over other possible solutions and/or over the prior art, whether or not a particular advantage is achieved by a given embodiment is not limiting of the disclosure. Thus, the following aspects, features, embodiments and advantages are merely illustrative and are not considered elements or limitations of the appended claims except where explicitly recited in a claim(s). Likewise, reference to “the disclosure” shall not be construed as a generalization of any inventive subject matter disclosed herein and shall not be considered to be an element or limitation of the appended claims except where explicitly recited in a claim(s).
(15) The present disclosure generally relates to a magnetic recording head comprising a read head. The read head comprises a first sensor disposed at a media facing surface (MFS) comprising at least one free layer, a second sensor disposed at the MFS comprising at least one free layer, a first spin generator spaced from the first sensor and recessed from the MFS, and a second spin generator spaced from the second sensor and recessed from the MFS. The first and second spin generators each individually comprises at least one spin orbit torque (SOT) layer. The SOT layer may comprise BiSb. The first and second sensors are configured to detect a read signal using a first voltage lead and a second voltage lead. The first and second spin generators are configured to inject spin current through non-magnetic layers to the first and second sensors using a plurality of current leads.
(16) FIG. **1** is a schematic illustration of a magnetic recording device **100**, according to one implementation. The magnetic recording device **100** includes a magnetic recording head, such as a write head. The magnetic recording device **100** is a magnetic media drive, such as a hard disk drive

(HDD). Such magnetic media drives may be a single drive/device or include multiple drives/devices. For the ease of illustration, a single disk drive is shown as the magnetic recording device **100** in the implementation illustrated in FIG. **1**. The magnet recording device **100** (e.g., a disk drive) includes at least one rotatable magnetic disk **112** supported on a spindle **114** and rotated by a drive motor **118**. The magnetic recording on each rotatable magnetic disk **112** is in the form of any suitable patterns of data tracks, such as annular patterns of concentric data tracks on the rotatable magnetic disk **112**.

(17) At least one slider **113** is positioned near the rotatable magnetic disk **112**. Each slider **113** supports a head assembly **121**. The head assembly **121** includes one or more magnetic recording heads (such as read/write heads), such as a write head including a spintronic device. As the rotatable magnetic disk **112** rotates, the slider **113** moves radially in and out over the disk surface **122** so that the head assembly **121** may access different tracks of the rotatable magnetic disk **112** where desired data are written. Each slider **113** is attached to an actuator arm **119** by way of a suspension **115**. The suspension **115** provides a slight spring force which biases the slider **113** toward the disk surface **122**. Each actuator arm **119** is attached to an actuator **127**. The actuator **127** as shown in FIG. **1** may be a voice coil motor (VCM). The VCM includes a coil movable within a fixed magnetic field, the direction and speed of the coil movements being controlled by the motor current signals supplied by a control unit **129**.

(18) The head assembly **121**, such as a write head of the head assembly **121**, includes a media facing surface (MFS) such as an air bearing surface (ABS) that faces the disk surface **122**. During operation of the magnetic recording device **100**, the rotation of the rotatable magnetic disk **112** generates an air or gas bearing between the slider **113** and the disk surface **122** which exerts an upward force or lift on the slider **113**. The air or gas bearing thus counter-balances the slight spring force of suspension **115** and supports the slider **113** off and slightly above the disk surface **122** by a small, substantially constant spacing during operation.

(19) The various components of the magnetic recording device **100** are controlled in operation by control signals generated by control unit **129**, such as access control signals and internal clock signals. The control unit **129** includes logic control circuits, storage means and a microprocessor. The control unit **129** generates control signals to control various system operations such as drive motor control signals on a line **123** and head position and seek control signals on a line **128**. The control signals on line **128** provide the desired current profiles to optimally move and position slider **113** to the desired data track on rotatable magnetic disk **112**. Write and read signals are communicated to and from the head assembly **121** by way of recording channel **125**. In one embodiment, which can be combined with other embodiments, the magnetic recording device **100** may further include a plurality of media, or disks, a plurality of actuators, and/or a plurality number of sliders.

(20) FIG. **2** is a schematic illustration of a cross sectional side view of a head assembly **200** facing the rotatable magnetic disk **112** shown in FIG. **1** or other magnetic storage medium, according to one implementation. The head assembly **200** may correspond to the head assembly **121** described in FIG. **1**. The head assembly **200** includes a MFS **212**, such as an ABS, facing the rotatable magnetic disk **112**. As shown in FIG. **2**, the rotatable magnetic disk **112** relatively moves in the direction indicated by the arrow **232** and the head assembly **200** relatively moves in the direction indicated by the arrow **233**.

(21) In one embodiment, which can be combined with other embodiments, the head assembly **200** includes a magnetic read head **211**. The magnetic read head **211** may include a sensing element **204** disposed between shields S**1** and S**2**. The sensing element **204** and related structures will be further described with respect to various embodiments shown in the disclosure. The magnetic fields of magnetized regions in the rotatable magnetic disk **112**, such as perpendicular recorded bits or longitudinal recorded bits, are detectable by the sensing element **204** as the recorded bits.

(22) The head assembly **200** includes a write head **210**. In one embodiment, which can be

combined with other embodiments, the write head **210** includes a main pole **220**, a leading shield **206**, a trailing shield (TS) **240**, and a spintronic device **250** disposed between the main pole **220** and the TS **240**. Each of the main pole **220**, the spintronic device **250**, the leading shield **206**, and the TS **240** has a front portion at the MFS.

(23) The main pole **220** includes a magnetic material, such as CoFe, CoFeNi, NiFe, or FeNiRe, other suitable magnetic materials. In one embodiment, which can be combined with other embodiments, the main pole **220** includes small grains of magnetic materials in a random texture, such as body-centered cubic (BCC) materials formed in a random texture. In one example, a random texture of the main pole **220** is formed by electrodeposition. The write head **210** includes a coil **218** around the main pole **220** that excites the main pole **220** to produce a writing magnetic field for affecting a magnetic recording medium of the rotatable magnetic disk **112**. The coil **218** may be a helical structure or one or more sets of pancake structures.

(24) In one embodiment, which can be combined with other embodiments, the main pole **220** includes a trailing taper **242** and a leading taper **244**. The trailing taper **242** extends from a location recessed from the MFS **212** to the MFS **212**. The leading taper **244** extends from a location recessed from the MFS **212** to the MFS **212**. The trailing taper **242** and the leading taper **244** may have the same degree or different degree of taper with respect to a longitudinal axis **260** of the main pole **220**. In one embodiment, which can be combined with other embodiments, the main pole **220** does not include the trailing taper **242** and the leading taper **244**. In such an embodiment, the main pole **220** includes a trailing side and a leading side in which the trailing side and the leading side are substantially parallel.

(25) The TS **240** includes a magnetic material, such as FeNi, or other suitable magnetic materials, serving as a second electrode and return pole for the main pole **220**. The leading shield **206** may provide electromagnetic shielding and is separated from the main pole **220** by a leading gap **254**.

(26) FIG. **3** illustrates a read head **300** based on NLSV. The read head **300** comprises a read sensor **340** disposed at the MFS adjacent to a media **301** and a spin generator **350** recessed from the MFS. The read sensor **340** and the spin generator **350** are each individually disposed between a first shield (not shown) and the second shield (not shown), and are separated, with the read sensor **340** being exposed to MFS and the spin generator **350** been recessed from MFS. The read sensor **340** and the spin generator **350** are also spaced from each other by some finite distance. A non-magnetic layer **306** is disposed adjacent to each of the first sensor **340** and the spin generator **350**. The non-magnetic layer **306** may comprise Cu or Al, for example.

(27) The read sensor **340** comprises a first free layer **344** and a first tunnel barrier layer **342** disposed between the first free layer **344** and the non-magnetic layer **306**. Voltage leads are connected to the non-magnetic layer **306** and the first free layer **344** for read signal detection through the read sensor **340**.

(28) The spin generator **350** comprises a reference layer **336**, a second tunnel barrier layer **332** disposed on the reference layer **336**, and a second free layer **334** disposed between the non-magnetic layer **306** and the second tunnel barrier layer **332**. The spin generator **350** is spaced a distance of about up to a spin diffusion length of the non-magnetic layer **306** from the read sensor **340**. For instance, if the non-magnetic layer **306** is made of Cu, the distance can be up to about 100 nm to about 300 nm. Current leads are connected to the non-magnetic layer **306** and the reference layer **336** for spin injection. During operation, when an electrical current is applied and applied through the spin generator **350**, a spin polarized current travels from the spin generator **350** down through the non-magnetic layer **306** to the read sensor **340**. The spin injection from the spin generator **350** allows the spin current to flow to the read sensor **340**, enabling the signal detection from the first free layer **344**'s magnetization changes when reading data from recording disk.

(29) However, since spin generator **350** comprises a reference layer **336** having a fixed magnetization direction, the spin injection efficiency is limited by the spin polarization of the reference layer **336** which cannot be bigger than 1. As such, the reader signal output of the read

head **300** is limited.

(30) FIGS. **4**A-**4**B illustrate a read head **400**, according to one embodiment. FIG. **4**A illustrates a cross-sectional view of the read head **400** and FIG. **4**B illustrates a side view of the read head **400**. The read head **400** may be within the read head **211** of FIG. **2**. The read head **400** may be a part of the magnetic recording device **100** of FIG. **1**. In the side view of FIG. **4**B, the first shield (S**1**) **402** and the second shield (S**2**) **404** are not shown for clarity purposes. The first shield **402** may be the S**1** of FIG. **2**, and the second shield **404** may be the S**2** of FIG. **2**. The read head **400** may be a two dimensional magnetic recording (TDMR) read head.

(31) The read head **400** comprises a read sensor **440** disposed at the MFS and a SOT spin generator **450** recessed from the MFS. The read sensor **440** is disposed between the first shield **402** and the second shield **404**. A non-magnetic layer **406** is disposed between the second shield **404** and each of the first and spin generators **440**, **450**. The non-magnetic layer **406** extends from the MFS to a back edge of the spin generator **450**. The non-magnetic layer **406** has a height **452** less than, or closer to, the spin diffusion length of the non-magnetic layer **406**, generally of about 100 nm to about 300 nm, depending on material used in the non-magnetic layer **406**. The non-magnetic layer **406** may comprise a material having a long spin diffusion length, such as Cu, Al, or some other 2D materials, such as graphene, and other van der Waals material, like MoS.sub.2, HfS.sub.2, etc. The read sensor **440** is spaced a distance **444** equal to about the height of the non-magnetic layer **406** from the spin generator **450**.

(32) In some embodiments, the non-magnetic layer **406** may have a greater height than the first and second shields **402**, **404**, or a smaller height than the first and second shields **402**, **404** (the configuration shown in FIG. **4**A). The non-magnetic layer **406** has a height **452** of less than, or closer to, the spin diffusion length of the non-magnetic layer **406**, generally of about 100 nm to about 300 nm, depending on material used in the non-magnetic layer **406**. In addition, the first and second shields **402** and **404** may have different dimensions (different heights from MFS).

(33) Depending on (1) the respective heights of the first shield **402** and the second shield **404**, and (2) the spin diffusion characteristics of the non-magnetic layer **406** (which influences the recess spacing of the spin generator **450**), the spin generator **450** may or may not be between the two shields. While FIGS. **4**A-**9**B show one or more spin generator between the shields and the shields are relatively similar or the same in dimensions, the disclosure is not so limited and there are other embodiments where the spin generator **450** may not be between the two shields. In those embodiments, a portion of the non-magnetic layer **406** that overlaps with the sensor could be between the two shields, but another portion of the non-magnetic layer **406** that overlaps with the spin generator **450** may not be.

(34) The read sensor **440** comprises at least one free layer **418** disposed adjacent to the first shield **402** and at least one tunnel barrier layer **442** disposed between the free layer **418** and the non-magnetic layer **406**. The free layer **418** may comprise a magnetic material such as one or more layers of Co—Fe, Co—Fe—B, Pt, or a Heusler alloy. The tunnel barrier layer **442** may comprise MgO or AlOx, for example, where x is a numeral greater than 1. Voltage leads are connected to the first and second shields **402**, **404** for read signal detection through the read sensor **440**. While only one free layer **418** and one tunnel barrier layer **442** are shown, the read sensor **440** may be a dual free layer (DFL) sensor, as discussed further below.

(35) The spin generator **450** is a multilayer structure. The SOT spin generator **450**, or the recessed read sensor **450**, may be any of the read sensors **500-800** discussed below in FIGS. **5**A-**8**B. Electrical current leads are fed into the spin generator **450** for spin injection. During operation, when current is applied to the read head **400** inside of the SOT spin generator **450** plane, due to a spin Hall effect, there will be a spin current travels from the SOT spin generator **450** down through the non-magnetic layer **406** to the read sensor **440**. This spin current flowing through the read sensor **440** enables an electrical signal generation across the sensor **440** when the free layer **418** rotates its magnetization directions when reading data.

(36) As shown in FIG. **4**B, the non-magnetic layer **406** may have a triangular or trapezoidal shape such that the non-magnetic layer **406** is wider further away from the MFS and narrows as the non-magnetic layer **406** gets closer to the MFS. In such embodiments, the wider portion of the non-magnetic layer **406** has a width **454** of less than or equal to about 100 nm, and the narrower portion of the non-magnetic layer **406** has a width **456** of about 10 nm to about 20 nm, which is closer to a track width (TW) of the read sensor **440**. The free layer **418** of the sensor **440** may have a same width **456** as the narrower portion of the non-magnetic layer **406** when the non-magnetic layer **406** varies in width. The spin generator **450** is disposed on a wider part of the non-magnetic layer **406**. Side shields **448** are disposed on either side of the non-magnetic layer **406** and read sensor **440** at the MFS for the proper biasing of the free layer **418**.
(37) FIGS. **4**C-**4**D illustrate a read head **401**, according to one embodiment. FIG. **4**C illustrates a cross-sectional view of the read head **401** and FIG. **4**D illustrates a top view of the read head **401**. The read head **401** may be within the read head **211** of FIG. **2**. The read head **401** may be a part of the magnetic recording device **100** of FIG. **1**. In the top view of FIG. **4**D, the first shield (S**1**) **402** and the second shield (S**2**) **404** are not shown for clarity purposes.
(38) The read head **401** of FIGS. **4**C-**4**D is similar to the read head **400** of FIGS. **4**A-**4**B; however, the spin generator **450** serves the function of spin detection, and will be referred to herein as a spin detector **450** or spin detector layer **450**, and the current and voltage leads are reversed, resulting in the direction the generated spin current flows through the non-magnetic layer **406** being reversed. In the read head **401**, input current leads (I) are connected to the first shield **402** and to the second shield **404**, and voltage leads (V) are connected to the spin detector **450**. During operation, when a current is applied to read head **401**, a spin current (I.sub.s) from the FL **418** is generated and flows across the read sensor **440** into the non-magnetic layer **406**. The spin current travels from the read sensor **440** up through the non-magnetic layer **406** to the spin detector **450**. The vertical spin current flowing through the spin detector **450** enables an electrical signal/voltage generation across the side (cross-track direction) of the spin detector **450** due to the inverse spin Hall effect. When the free layer **418** rotates its magnetization directions when reading data, spin polarization of the spin current (I.sub.s) will change, resulting in an output voltage (V.sub.out) across the spin detector **450** reflecting the magnetic bit information on the recording track.
(39) Furthermore, when the current leads (I) are connected to the first shield **402** and to the second shield **404**, and voltage leads (V) are connected to the spin detector **450**, the non-magnetic layer **406** has a rectangular shape. In such embodiments, the non-magnetic layer **406** has a consistent narrow width **456** of about 10 nm to about 20 nm, which is closer to a track width (TW) of the read sensor **440**.
(40) FIGS. **5**A-**8**B illustrate various embodiments of read heads **500**, **501**, **600**, **601**, **700**, **800**, respectively, according to various embodiments. Each read head **500**, **600**, **700**, **800** may individually be the read head **400** of FIGS. **4**A-**4**B. Each read head **501**, **601** may individually be the read head **401** of FIGS. **4**C-**4**D. Each read head **500**, **501**, **600**, **601**, **700**, **800** may individually be within the read head **211** of FIG. **2**. Each read head **500**, **501**, **600**, **601**, **700**, **800** may individually be a part of the magnetic recording device **100** of FIG. **1**. Aspects of FIGS. **5**A-**8**B may not be shown to scale, such as the height of the first and second shields **402**, **404** in the x-direction. Furthermore, aspects of each read head **500**, **501**, **600**, **601**, **700**, **800** may be used in combination with one another and/or with the read head **400** of FIGS. **4**A-**4**B. As such, various layers and/or aspects of the read heads **400**, **401**, **500**, **501**, **600**, **601**, **700**, **800** may have consistent reference numerals.
(41) FIGS. **5**A-**5**B illustrate a TDMR read head **500** of a magnetic recording head, according to one embodiment. FIG. **5**A illustrates a cross-sectional view of the TDMR read head **500** and FIG. **5**B illustrates a MFS view of the TDMR read head **500**.
(42) The TDMR read head **500** comprises a first lower shield **402**, a first non-magnetic layer **506***a* disposed over the first lower shield **402**, a first, or lower, free layer (FL) read sensor **540***a* disposed

at the MFS on the first non-magnetic layer **506***a*, a first, or lower, a spin orbit torque (SOT) spin generator **550***a* recessed from the MFS on the first non-magnetic layer **506***a*, a first upper shield **404** disposed over the first FL read sensor **540***a* and the first spin generator **550***a*, and an insulating layer **514***a* disposed between the first FL sensor **540***a* and the first SOT generator **550***a*. A first notch **516***a* is disposed between the first lower shield **402** and the first non-magnetic layer **506***a* and is recessed from the MFS.

(43) The TDMR read head **500** further comprises an insulating layer **542** disposed on the first upper shield **404**, a second lower shield **502** disposed on the insulating layer **542**, a second non-magnetic layer **506***b* disposed over the second lower shield **502**, a second, or upper, FL read sensor **540***b* disposed at the MFS on the second non-magnetic layer **506***b*, a second, or upper, SOT spin generator **550***b* recessed from the MFS on the second non-magnetic layer **506***b*, a second upper shield **504** disposed over the second FL read sensor **540***b* and the second SOT spin generator **550***b*, and an insulating layer **514***b* disposed between the second FL sensor **540***b* and the second SOT generator **550***b*. A second notch **516***b* is disposed between the second lower shield **502** and the second non-magnetic layer **506***b* and recessed from the MFS.

(44) Each SOT layer **510***a*, **510***b* is spaced the distance **444** from the first free layer **418***a* and the second free layer **418***b*, respectively. In some embodiments, the first upper shield **404** and the second lower shield **502** are a single middle shield, and the insulating layer **542** is not included, like shown in the TDMR read head **800** of FIGS. **8**A-**8**B.

(45) The first notch **516***a* and the second notch **516***b* are each individually recessed a distance of about 0 nm to about 20 nm from the MFS. The first notch **516***a* and the second notch **516***b* each individually has a thickness in the y-direction of about 5 nm to about 20 nm and a height in the x-direction of about 5 nm to about 20 nm. The first lower shield **402**, first upper shield **404**, second lower shield **502**, second upper shield **504**, the first notch **516***a*, and the second notch **516***b* may each individually comprise a magnetic material similar to the soft bias material, such as NiFe, NiFe/CoFe laminates, NiFe/NiFeCr laminates, or NiFe/W laminates, for example ("/" as used here denotes separate layers in a multi-layer stack). A first dielectric layer **515***a* is disposed in front of the first notch **516***a* at the MFS and behind the first notch **516***a* recessed from the MFS, and a second dielectric layer **515***b* is disposed in front of the second notch **516***b* at the MFS and behind the second notch **516***b* recessed from the MFS. The first dielectric layer **515***a* spaces the first lower shield **402** from the first non-magnetic layer **506***a*, and the second dielectric layer **515***b* spaces the second lower shield **502** from the second non-magnetic layer **506***b*. The first dielectric layer **515***a* and the second dielectric layer **515***b* may each individually comprise SiN, SiO.sub.2, MgO, AlO, AlN, or combinations thereof.

(46) The first FL sensor **540***a* comprises a first tunnel barrier layer **520***a* disposed on the first non-magnetic layer **506***a*, a first free layer **418***a* disposed on the first tunnel barrier layer **520***a*, and a first cap layer **522***a* disposed on the first free layer **418***a* and in contact with the first upper shield **404**. A magnetization (M) direction of the first free layer **418***a* is in the −z-direction at quiescent state during operation. Similarly, the second FL sensor **540***b* comprises a second tunnel barrier layer **520***b* disposed on the second non-magnetic layer **506***b*, a second free layer **418***b* disposed on the second tunnel barrier layer **520***b*, and a second cap layer **522***b* disposed on the second free layer **418***b* and in contact with the second upper shield **504**. A magnetization (M) direction of the second free layer **418***b* in the −z-direction at quiescent state during operation, or parallel to the magnetization direction of the first free layer **418***a*. The first and second tunnel barrier layers **520***a*, **520***b* (see list of materials in FIG. **4**A tunnel barrier layer **442**), and the first and second cap layers **522***a*, **522***b*, may each individually be multilayer structures or single layer/material structures.

(47) The first SOT spin generator **550***a* comprises a first seed layer **508***a* disposed on the first non-magnetic layer **506***a*, a first SOT layer **510***a* (which may be referred to herein as a first BiSb layer **510***a*) disposed on the first seed layer **508***a*, and a first cap layer **512***a* disposed on the first SOT layer **510***a*. The second SOT spin generator **550***b* comprises a second seed layer **508***b* disposed on

the second non-magnetic layer **506***b*, a second SOT layer **510***b* (which may be referred to herein as a second BiSb layer **510***b*) disposed on the second seed layer **508***b*, and a second cap layer **512***b* disposed on the second SOT layer **510***b*. As shown in FIG. **5**B, the first SOT layer **510***a* has a greater overall size in the z-direction than the first free layer **418***a*, and the second SOT layer **510***b* has a greater overall size in the z-direction than the second free layer **418***b*. The first SOT layer **510***a* and the second SOT layer **510***b* may have the same dimensions, and the first free layer **418***a* and the second free layer **418***b* may have the same dimensions.
(48) The first and second cap layers **512***a*, **512***b* may each individually comprise a non-magnetic, high resistivity material, such as Ru, NiFeGe, or some oxide materials, for example. The first and second cap layers **512***a*, **512***b* may be multilayer structures or comprise a single layer/material. In some embodiments, the insulating layers **514***a*, **514***b* each extend over the first and second cap layers **512***a*, **512***b*, respectively, and behind the first and second SOT sensors **550***a*, **550***b*, like shown in FIG. **5**A. The insulating layers **514***a*, **514***b*, **542** may each individually comprise SiN, SiO.sub.2, MgO, AlO, AlN, MgTiOx, or combinations thereof.
(49) A first negative voltage lead (V.sub.1−) is connected to the first lower shield **402** and a first positive voltage lead (V.sub.1+) is connected to the first upper shield **404** for read signal detection through the first FL sensor **540***a*. A second negative voltage lead (V.sub.2−) is connected to the second lower shield **502** and a second positive voltage lead (V.sub.2+) is connected to the second upper shield **504** for read detection across the second FL sensor **540***b*.
(50) As shown in the MFS view of FIG. **5**B, the TDMR read head **500** further comprises a first soft bias (SB) side shield **524***a* and a second SB side shield **526***a* each adjacent to the first FL sensor **540***a* at the MFS. A third SB side shield **524***b* and a fourth SB side shield **526***b* are disposed adjacent to the second FL sensor **540***b* at the MFS. The first, second, third, and fourth SB side shields **524***a*, **526***a*, **524***b*, **526***b* may each individually comprise NiFe, CoFe, and other magnetic materials for proper free layer biasing across track (i.e., the z-direction in FIG. **5**B). A first positive current lead (I.sub.1+) and a first negative current lead (I.sub.1−) are each connected to the first SOT layer **510***a*. A first electrical current (I.sub.1) travels through the first SOT layer **510***a* into the page of FIG. **5**A, or the −z-direction. A second positive current lead (I.sub.2+) and a second negative current lead (I.sub.2−) are each conned to the second SOT layer **510***b*. A second electrical current (I.sub.2) travels through the second SOT layer **510***b* into the page of FIG. **5**A, or the −z-direction.
(51) During operation, when electrical currents are applied to the TDMR read head **500** as indicated by I.sub.1 and I.sub.2, a first spin current travels from the first SOT spin generator **550***a* down through the first non-magnetic layer **506***a* to the first FL sensor **540***a*, and a second spin current travels from the second SOT spin generator **550***b* down through the second non-magnetic layer **506***b* to the second FL sensor **540***b*. The spin transport allows the spin current to flow to the first and second FL sensors **540***a*, **540***b*, enabling an electrical signal generation across the first sensor **540***a* and the second sensor **540***b* when the first and second free layers **418***a*, **418***b* rotate magnetization directions when reading data.
(52) The first and second SOT layers **510***a*, **510***b* each individually comprises heavy metal with strong spin orbital coupling, such as Ta, Pt, W, Hf, etc. The first and second SOT layers **510***a*, **510***b* can individually comprise a topological insulator material, such as BiSe, WTe, YBIOPt, or BiSb. The first and second SOT layers **510***a*, **510***b* may each individually comprise BiSb in a (012) orientation. The first and second SOT layers **510***a*, **510***b* may each individually comprise undoped BiSb or doped BiSbX, where the dopant is less than about 10 at. %, and where X is extracted from elements which don't readily interact with Bi, such as B, N, Al, Si, Ti, V, Cr, Fe, Ni, Cu, Ge, Y, Zr, Ru, Mo, Ag, Hf, Ta, W, or Ir, or in alloy combinations with one or more of aforementioned elements, like CuAg, CuNi, RuGe, etc. More generally, some of the listed dopants may be used with other topological insulator materials other than BiSb. Each SOT layer **510***a*, **510***b* has a thickness **558** in the y-direction (e.g., a down-track direction) of about 5 nm to about 20, a height

560 in the x-direction (e.g., a stripe height) of about 100 nm to about 1 μm, and a width **562** (e.g., a cross-track direction) in the z-direction of less than about 100 nm.

(53) Numerous materials may be utilized as the first and second seed layers **508***a*, **508***b* to provide a textured SOT layer **510** (012) growth using either textures (100) or as amorphous seed layers. The first and second seed layers **508***a*, **508***b* may each be multilayer structures or comprise a single layer/material. One group of materials, Group A, includes body centered cubic (BCC) such as V, V.sub.3Al, Mn.sub.3Al, Nb, Mo, W, Ta, WTi.sub.50, NiAl, RhAl, or in alloy combinations of these materials with similar lattice parameters, or a BCC material used in combination with (100) textured layers such as Cr (heated at about 250° C. or larger), RuAl, IrAl, CoAl, B2 phases, NiAl—B2 phase, CrMo(Mo of about 20-50 at. %)-A2, or B2 phase, A2 CrX(X is about at. 10%, heated at about 250° C. or larger, and is selected from Ru, Ti, W, and Mo).

(54) The Group A materials provide texturing for subsequent layers and may be referred to as a MgO (100) texturing layer stack. Generally speaking, depositing a high boron (B) affinity elements or alloys such as alloys of tantalum (Ta), tungsten (W), or titanium (Ti) and then depositing a magnetic layer containing boron (B), an amorphous magnetic layer material such as cobalt iron boron (CoFeB) or cobalt boron (CoB) is formed, but the boron would be pulled out leaving a magnetic cobalt iron or cobalt layer behind. On top of that layer, MgO could be formed with a (100) texture. Other manners to make MgO (100) can include depositing heated chromium or ruthenium aluminum grow in a textured fashion. MgO (100) texturing layer stacks can also be made by depositing thin MgO on a magnetic bilayer of Co, or CoFe on a magnetic boron alloy of CoFeB or CoB deposited on a thin high B gettering alloy seed layers of Hf, Ta, W, Ti, or alloys containing these elements.

(55) Another group of materials, Group B, includes face centered cubic (FCC) oxide materials including FeO, CoO, NiO, ZrO, MgO, TiO, MgTiO, and MnO. Another group of materials, Group C, includes FCC nitrides and carbides including ScN, TiN, NbN, ZrN, HIN, TaN, VN, CrN, ScC, TiC, NbC, ZrC, HfC, TaC, WC, VC, and W.sub.0.8Zr.sub.0.2C. Group C materials can be deposited as amorphous to nanocrystalline thin films depending upon deposition conditions. Resistivities of Group C materials are >100-200 micro ohm-cm.

(56) Another groups of materials, Group D, includes nonmagnetic Heusler materials including FezVAl, Cr.sub.2CoAl, CoTiSb, Mn.sub.2VSi, V.sub.2Al, [Mn.sub.0.5Co.sub.0.5].sub.2VAl, [Mn.sub.0.75Co.sub.0.25].sub.2VSi, CoMnNbAl, CoZrFeAl, and Ti.sub.2MnAl. Another group of materials, Group E, are crystalline high polarization layers using magnetic alloys or Heusler alloys that have large spin polarizations and do not readily mix with a spin Hall layer including: Co.sub.2MnSb, CoFeX, NiFeX, (where X is one or more of Si, Al, Mn, and Ge) CoFe, NiFe, Co.sub.2MnGe, CoMnSb, NiMnSb, Co.sub.2FeGe, Co.sub.2MnSn, and Co.sub.2MnFeGe. Another group of materials, Group F, includes amorphous nonmagnetic high resistive electrical shunt block layers that do not promote strong (012) BiSbX texture or growth. They include SiO.sub.2, Al.sub.2O.sub.3, SiN, AlN, SiC, SiCrOx, Nix, Fex, and Cox, where X can be one or more of these elements including Fe, Co, Ni, Ta, Hf, W, Ir, Pt, Ti, Zr, N, Ru, Ge and/or B.

(57) Another group of materials, Group G, includes any metal amorphous or ceramic amorphous material with the nearest neighbor x-ray diffraction peak in the 2.19 Å to 2.02 Å d-spacing range. Such materials include nonmagnetic and magnetic materials from Group A, D, or E which are laminated or alloyed with one or more elements of: Cu, Ag, Ge, Al, Mg, Si, Mn, Ni, Co, Mo, Zr, Y, Bi, Hf, Ta, W, Ir, Pt, Ti, or B. They form effectively nonmagnetic amorphous layers which produce amorphous materials or start with amorphous materials like a-Ge, and a-NiPX alloys (where X is one of Ru, Rh, Y, Zr, Mo, Hf, Ta, W, Re, Pt, or Ir), etc. Group G also may include amorphous/nanocrystalline alloys with a-Ge, a-NiP and one or more elements of: Cu, Ag, Ge, Al, Mg, Si, Mn, Ni, Co, Mo, Zr, Y, Bi, Hf, Ta, W, Ir, Pt, Ti, or B to promote a strong (012) BiSb texture.

(58) Yet another group of materials, Group H, are high perpendicular magnetic anisotropy (PMA)

materials that can be amorphous or crystalline materials. Amorphous rare-earth transition metals (RE-TM) that have high PMA like TbFeCo, TbFeB, Nd, Pr, Sm(Fe, Co) B or heavy metals like CoZrTaB can be used. This multilayer polycrystalline stacks of Co/Pt, Co/Pd, CoFe/Pt, Co/Tb, or CoFe/Tb ("/" denoting layer separation), or single layer PMA materials like CoPt, CoPtCr, CoFePt, and FePt with high Ku can be used with an amorphous high polarizing layer next to the spin Hall layer for (012) texture growth.

(59) FIGS. **5**C-**5**D illustrate a read head **501** of a magnetic recording head, according to one embodiment. FIG. **5**C illustrates a cross-sectional view of the read head **501** and FIG. **5**D illustrates a MFS view of the read head **501**. In the read head **501**, the non-magnetic layer **406** may be rectangular in shape, like shown and discussed above in FIGS. **4**A-**4**D.

(60) The read head **501** is similar to the read head **500** of FIGS. **5**A-**5**B; however, the current and voltage leads are reversed. In the read head **501**, a first negative current lead (I.sub.1−) is connected to the first lower shield **402** and a first positive current lead (I.sub.1+) is connected to the first upper shield **404** for spin injection, enabling an electrical signal generation across the first sensor **540***a* when the first free layer **418***a* rotates magnetization directions when reading data. A second negative current lead (I.sub.2−) is connected to the second lower shield **502** and a second positive current lead (I.sub.2+) is connected to the second upper shield **504** for spin injection, enabling an electrical signal generation across the second sensor **540***b* when the second free layer **418***b* rotates magnetization directions when reading data. A first positive voltage lead (V.sub.1+) and a first negative voltage lead (V.sub.1−) are each connected to the first SOT layer **510***a* read signal detection of the first FL sensor **540***a*. A second positive voltage lead (V.sub.2+) and a second negative voltage lead (V.sub.2−) are each conned to the second SOT layer **510***b* for read detection of the second FL sensor **540***b*.

(61) During operation, when the input current is applied to the TDMR read head **501** using the first and second positive current leads and the first and second negative current leads, a first spin current is generated from the first FL sensor **540***a* and flows through the first non-magnetic layer **506***a* to the SOT spin generator **550***a* where the first spin current flows out-of-plane (i.e., in the y-direction), and a second spin current travels is generated the second FL sensor **540***b* and flows through the second non-magnetic layer **506***b* to the second SOT spin generator **550***b* where the second spin current flows out-of-plane (i.e., in the y-direction). The vertical spin current flowing into the SOT layers **510***a*, **510***b* enables an electrical signal generation across the SOT layers **510***a*, **510***b* via the inverse spin Hall effect when the free layers **418***a*, **418***b* each rotates its magnetization directions when reading data. Because the two split spin currents flow opposite in the y-direction, the generated signal polarity will be opposite as well, as shown by the voltage leads V.sub.2+/V.sub.2− and V.sub.1+/V.sub.1− in FIG. **5**D. The first positive voltage lead (V.sub.1+) and the second negative voltage lead (V.sub.2−) are connected together such that the final signal read out occurs at the first negative voltage lead (V.sub.1−) and the second positive voltage lead (V.sub.2+). FIGS. **6**A-**6**B illustrate a TDMR read head **600** of a magnetic recording head, according to another embodiment. FIG. **6**A illustrates a cross-sectional view of the TDMR read head **600** and FIG. **6**B illustrates a MFS view of the TDMR read head **600**. The TDMR read head **600** is similar to the TDMR read head **500** of FIGS. **5**A-**5**B; however, the first SOT spin generator **550***a* and the second SOT spin generator **550***b* each individually comprise two SOT layers.

(62) In the TDMR read head **600**, a first shield notch **602***a* is disposed between the first lower shield **402** and the first notch **516***a* of the first non-magnetic layer **506***a*. The first shield notch **602***a* is disposed at the MFS, and may comprise the same materials as the first lower shield **402**. A third cap layer **612***a* is disposed over the first lower shield **402**, recessed from the MFS. A third SOT layer **610***a* is disposed on the third cap layer **612***a*, and a third seed layer **608***a* is disposed on the third SOT layer **610***a*. The first non-magnetic layer **506***a* is disposed over the third seed layer **608***a*. A third insulating layer **614***a* is disposed between the first shield notch **602***a* and the third cap layer **612***a*, the third SOT layer **610***a*, and the third seed layer **608***a*.

(63) The third insulating layer **614***a* may comprise the same materials as the first insulating layer **514***a*, the third cap layer **612***a* may comprise the same materials as the first cap layer **512***a*, the third SOT layer **610***a* may comprise the same materials as the first SOT layer **510***a*, and the third seed layer **608***a* may comprise the same materials as the first seed layer **508***a*. The third insulating layer **614***a* may extend between the third cap layer **612***a* and the first lower shield **402** to the area behind the third cap layer **612***a*, the third SOT layer **610***a*, and the third seed layer **608***a*. The first shield notch **602***a* may comprise the same material as the first notch **516***a* and/or the second notch **516***b*.
(64) Similarly, a second shield notch **602***b* is disposed between the second lower shield **502** and the second notch **516***b* of the second non-magnetic layer **506***b*. The second shield notch **602***b* is disposed at the MFS. A fourth cap layer **612***b* is disposed over the second lower shield **502**, recessed from the MFS. A fourth SOT layer **610***b* is disposed on the fourth cap layer **612***b*, and a fourth seed layer **608***b* is disposed on the fourth SOT layer **610***b*. The second non-magnetic layer **506***b* is disposed over the fourth seed layer **608***b*. A fourth insulating layer **614***b* is disposed between the second shield notch **602***b* and the fourth cap layer **612***b*, the fourth SOT layer **610***b*, and the fourth seed layer **608***b*.
(65) The fourth insulating layer **614***b* may comprise the same materials as the second insulating layer **514***b*, the fourth cap layer **612***b* may comprise the same materials as the second cap layer **512***b*, the fourth SOT layer **610***b* may comprise the same materials as the second SOT layer **510***b*, and the fourth seed layer **608***b* may comprise the same materials as the second seed layer **508***b*. The third SOT layer **610***a* and the fourth SOT layer **610***b* may each individually have the same dimensions as the first SOT layer **510***a* and the second SOT layer **510***b*. The fourth insulating layer **614***b* may extend between the fourth cap layer **612***b* and the second lower shield **502** to the area behind the fourth cap layer **612***b*, the fourth SOT layer **610***b*, and the fourth seed layer **608***b*. The second shield notch **602***b* may comprise the same material as the first shield notch **602***a*.
(66) Each SOT layer **510***a*, **510***b*, **610***a*, **610***b* is spaced the distance **444** from the first free layer **418***a* and the second free layer **418***b*, respectively. In some embodiments, the first upper shield **404** and the second lower shield **502** are one middle shield, and the insulating layer **542** is not included, like shown in the TDMR read head **800** of FIGS. **8**A-**8**B.
(67) A first negative voltage lead (V.sub.1−) is connected to the first shield notch **602***a* and a first positive voltage lead (V.sub.1+) is connected to the first upper shield **404** for read signal detection through the first FL sensor **640***a*. A second negative voltage lead (V.sub.2−) is connected to the second shield notch **602***b* and a second positive voltage lead (V.sub.2+) is connected to the second upper shield **504** for read detection through the second FL sensor **640***b*.
(68) As shown in the MFS view of FIG. **6**B, the TDMR read head **600** further comprises a first soft bias (SB) side shield **524***a* and a second SB side shield **526***a* each adjacent to the first FL sensor **640***a* at the MFS. A third SB side shield **524***b* and a fourth SB side shield **526***b* are disposed adjacent to the second FL sensor **640***b* at the MFS. The first, second, third, and fourth SB side shields **524***a*, **526***a*, **524***b*, **526***b* may each individually comprise NiFe, CoFe, and other magnetic materials for proper free layer biasing across track. The first shield notch **602***a* is disposed in front of the third SOT layer **610***a* at the MFS, and the second shield notch **602***b* is disposed in front of the fourth SOT layer **610***b* at the MFS.
(69) A first positive current lead (I.sub.1+) and a first negative current lead (I.sub.1−) are each connected to the first SOT layer **510***a*. A first electrical current (I.sub.1) travels through the first SOT layer **510***a* into the page of FIG. **6**A, or the −z-direction. A second positive current lead (I.sub.2+) and a second negative current lead (I.sub.2−) are each connected to the second SOT layer **510***b*. A second electrical current (I.sub.2) travels through the second SOT layer **510***b* into the page of FIG. **6**A, or the −z-direction. A third positive current lead (I.sub.3+) and a third negative current lead (I.sub.3−) are each connected to the third SOT layer **610***a*. A third electrical current (I.sub.3) travels through the third SOT layer **610***a* out of the page of FIG. **6**A, or in the z-direction.

A fourth positive current lead (I.sub.4+) and a fourth negative current lead (I.sub.4−) are each connected to the fourth SOT layer **610***b*. A fourth electrical current (I.sub.4) travels through the fourth SOT layer **610***b* out of the page of FIG. **6**A, or in the z-direction.
(70) Thus, from the first lower shield **402** to the second upper shield **504**, the third SOT layer **610***a* has an electrical current traveling in the z-direction, the first SOT layer **510***a* has electrical current traveling in the −z-direction, the fourth SOT layer **610***b* has an electrical current traveling in the z-direction, and the second SOT layer **510***b* has an electrical current traveling in the −z-direction. As such, the first, second, third, and fourth SOT layers **510***a*, **510***b*, **610***a*, **610***b* have alternating electrical current directions.
(71) Moreover, the first positive current lead (I.sub.1+) is aligned with, or disposed adjacent to, to the third negative current lead (I.sub.3−), and the second positive current lead (I.sub.2+) is aligned with, or disposed adjacent to, to the fourth negative current lead (I.sub.4−). As such, the first negative current lead (I.sub.1−) is aligned with, or disposed adjacent to, to the third positive current lead (I.sub.3+), and the second negative current lead (I.sub.2−) is aligned with, or disposed adjacent to, to the fourth positive current lead (I.sub.4+). Thus, the positive and negative current leads are alternating.
(72) During operation, when the electrical currents described above are applied to the TDMR read head **600**, a first spin current travels from the first SOT spin generator **650***a* down through the first non-magnetic layer **506***a* to the first FL sensor **640***a*, and a second spin current travels from the second SOT spin generator **650***b* down through the second non-magnetic layer **506***b* to the second FL sensor **640***b*. Having two SOT layers **510***b*, **610***b* on opposite sides of the second non-magnetic layer **506***b* with opposite electrical currents can double the amount of the first spin current generated. Similarly, having two SOT layers **510***a*, **610***a* on opposite sides of the first non-magnetic layer **506***a* with opposite electrical currents can double the amount of the second spin current generated. Those spin injections allow the spin current to flow to the first and second FL sensors **640***a*, **640***b*, enabling an electrical signal generation across the first sensor **640***a* and the second sensor **640***b* when the first and second free layers **418***a*, **418***b* rotate magnetization directions when reading data.
(73) FIGS. **6**C-**6**D illustrate a read head **601** of a magnetic recording head, according to one embodiment. FIG. **6**C illustrates a cross-sectional view of the read head **601** and FIG. **6**D illustrates a MFS view of the read head **601**. In the read head **601**, the non-magnetic layer **406** may be rectangular in shape, like shown and discussed above in FIGS. **4**A-**4**D.
(74) The read head **601** is similar to the read head **600** of FIGS. **6**A-**6**B; however, the current and voltage leads are reversed. In the read head **601**, a first negative current lead (I.sub.1−) is connected to the first shield notch **602***a* and a first positive current lead (I.sub.1+) is connected to the first upper shield **404** for spin injection, enabling an electrical signal generation across the first sensor **540***a* when the first free layer **418***a* rotates magnetization directions when reading data. A second negative current lead (I.sub.2−) is connected to the second shield notch **602***b* and a second positive current lead (I.sub.2+) is connected to the second upper shield **504** for spin injection, enabling an electrical signal generation across the second sensor **540***b* when the second free layer **418***b* rotates magnetization directions when reading data.
(75) A first positive voltage lead (V.sub.1+) and a first negative voltage lead (V.sub.1−) are each connected to the first SOT layer **510***a* and a third negative voltage lead (V.sub.3−), and a third positive voltage lead (V.sub.3+) are each connected to the third SOT layer **610***a*, for read signal detection of the first FL sensor **640***a*. The first negative voltage lead (V.sub.1−) and the third positive voltage lead (V.sub.3+) are connected together such that the final signal read out occurs at the first positive voltage lead (V.sub.1+) and the third negative voltage lead (V.sub.3−). A second positive voltage lead (V.sub.2+) and a second negative voltage lead (V.sub.2−) are each connected to the second SOT layer **510***b* and a fourth negative voltage lead (V.sub.4−), and a fourth positive voltage lead (V.sub.4+) are each connected to the fourth SOT layer **610***b*, for read signal detection

of the second FL sensor **640***b*. The second negative voltage lead (V.sub.2−) and the fourth positive voltage lead (V.sub.4+) are connected together such that the final signal read out occurs at the second positive voltage lead (V.sub.2+) and the fourth negative voltage lead (V.sub.4−).

(76) During operation, when the input current is applied to the read head **601** using the first and second positive current leads and the first and second negative current leads, a first spin current (I.sub.s1) is generated from the first FL sensor **640***a* and flows through the first non-magnetic layer **506***a* to the SOT spin generator **550***a* where the first spin current flows out-of-plane (i.e., in the y-direction), and a second spin current (I.sub.s2) travels is generated the second FL sensor **540***b* and flows through the second non-magnetic layer **506***b* to the second SOT spin generator **550***b* where the second spin current flows out-of-plane (i.e., in the y-direction). Two inverse spin Hall effect voltages are then induced separately on the first and third SOT layers **510***a*, **610***a*, and two inverse spin Hall effect voltages are then induced separately on the second and fourth SOT layers **510***b*, **610***b*. The vertical spin currents flowing into the first and third SOT layers **510***a*, **610***a* in opposite directions, and into the second and fourth SOT layers **510***b*, **610***b* in opposite directions, enable an electrical signal generation across the SOT layers **510***a*, **510***b*, **610***a*, **610***b* via the inverse spin Hall effect when the free layers **418***a*, **418***b* each rotates its magnetization directions when reading data. Because the two split spin currents flow opposite in the y-direction, the generated signal polarity will be opposite as well, as shown by the voltage leads V.sub.1+/V.sub.1−, V.sub.2+/V.sub.2−, V.sub.3+/V.sub.3−, and V.sub.4+/V.sub.4− in FIG. **6**D.

(77) FIGS. **7**A-**7**B illustrate a TDMR read head **700** of a magnetic recording head, according to yet another embodiment. FIG. **7**A illustrates a cross-sectional view of the TDMR read head **700** and FIG. **7**B illustrates a MFS view of the TDMR read head **700**. The TDMR read head **700** is similar to the read head **600** of FIGS. **6**A-**6**B; however, the first and second sensors **740***a*, **740***b* are DFL sensors. Further, the read head **700** does not comprise the first shield notch **602***a*, the second shield notch **602***b*, the first notch **516***a*, or the second notch **516***b*.

(78) In the TDMR read head **700**, the first FL sensor **740***a* and the second FL sensor **740***b* are each dual free layer (DFL) sensors. A third free layer **718***a* is disposed between the first lower shield **402** and the first non-magnetic layer **506***a*, and a fourth free layer **718***b* is disposed between the second lower shield **502** and the second non-magnetic layer **506***b*. A third tunnel barrier layer **722***a* is disposed between the third free layer **718***a* and the first non-magnetic layer **506***a*, and a fourth tunnel barrier layer **722***b* is disposed between the fourth free layer **718***b* and the second non-magnetic layer **506***b*. A third cap layer **720***a* is disposed between the first lower shield **402** and the third free layer **718***a*, and a fourth cap layer **720***b* is disposed between the fourth free layer **718***b* and the second lower shield **502**. The third and fourth free layers **718***a*, **718***b* may each individually comprise the same materials as the first and/or second free layers **418***a*, **418***b*. The third and fourth tunnel barrier layers **722***a*, **722***b* may each individually comprise the same materials as the first and second tunnel barrier layers **520***a*, **520***b*. The third and fourth cap layers **720***a*, **720***b* may each individually comprise the same materials as the first and cap layers **522***a*, **522***b*. Each SOT layer **510***a*, **510***b*, **610***a*, **610***b* is spaced the distance **444** from the first free layer **418***a*, the second free layer **418***b*, the third free layer **718***a*, and the fourth free layer **718***b*, respectively.

(79) The TDMR read head **700** further comprises a fifth SB side shield **724***a* and a sixth SB side shield **726***a* disposed adjacent to the third free layer **718***a* at the MFS, and a seventh SB side shield **724***b* and an eighth SB side shield **726***b* disposed adjacent to the fourth free layer **718***b* at the MFS. A first Ru layer **827***a* is disposed between and in contact with the first SB side shield **524***a* and the fifth SB side shield **724***a*. A second Ru layer **827***b* is disposed between and in contact with the second SB side shield **526***a* and the sixth SB side shield **726***a*. A third Ru layer **827***c* is disposed between and in contact with the third SB side shield **524***b* and the seventh SB side shield **724***b*. A fourth Ru layer **827***d* is disposed between and in contact with the fourth SB side shield **526***b* and the eighth SB side shield **726***b*. The thickness of the Ru of the Ru layers **827***a*-**827***d* is adjusted so that magnetization of the SB side shields **724***a*, **724***b*, **524***a*, **524***b* are anti-parallel to each other

along the cross track direction.
(80) The first free layer **418***a* has a magnetization direction in the −z-direction, the second free layer **418***b* has a magnetization direction in the −z-direction, the third free layer **718***a* has a magnetization direction in the z-direction, and the fourth free layer **718***b* has a magnetization direction in the z-direction. Thus, from the first lower shield **402** to the second upper shield **504**, the first, second, third, and fourth free layers **418***a*, **418***b*, **718***a*, **718***b* have alternating magnetization directions.
(81) A first positive current lead (I.sub.1+) and a first negative current lead (I.sub.1−) are each connected to the first SOT layer **510***a*. A first electrical current (I.sub.1) travels through the first SOT layer **510***a* into the page of FIG. **7**A, or the −z-direction. A second positive current lead (I.sub.2+) and a second negative current lead (I.sub.2−) are each connected to the second SOT layer **510***b*. A second electrical current (I.sub.2) travels through the second SOT layer **510***b* into the page of FIG. **7**A, or the −z-direction. A third positive current lead (I.sub.3+) and a third negative current lead (I.sub.3−) are each connected to the third SOT layer **610***a*. A third electrical current (I.sub.3) travels through the third SOT layer **610***a* out of the page of FIG. **7**A, or in the z-direction. A fourth positive current lead (I.sub.4+) and a fourth negative current lead (I.sub.4) are each connected to the fourth SOT layer **610***b*. A fourth electrical current (I.sub.4) travels through the fourth SOT layer **610***b* out of the page of FIG. **7**A, or in the z-direction.
(82) Thus, from the first lower shield **402** to the second upper shield **504**, the third SOT layer **610***a* has an electrical current traveling in the z-direction, the first SOT layer **510***a* has an electrical current traveling in the −z-direction, the fourth SOT layer **610***b* has an electrical current traveling in the z-direction, and the second SOT layer **510***b* has an electrical current traveling in the −z-direction. As such, the first, second, third, and fourth SOT layers **510***a*, **510***b*, **610***a*, **610***b* have alternating current directions.
(83) Moreover, the first positive current lead (I.sub.1+) is aligned with, or disposed adjacent to, to the third negative current lead (I.sub.3−), and the second positive current lead (I.sub.2+) is aligned with, or disposed adjacent to, to the fourth negative current lead (I.sub.4−). As such, the first negative current lead (I.sub.1−) is aligned with, or disposed adjacent to, to the third positive current lead (I.sub.3+), and the second negative current lead (I.sub.2−) is aligned with, or disposed adjacent to, to the fourth positive current lead (I.sub.4+). Thus, the positive and negative current leads are alternating.
(84) During operation, when the currents described above are applied to the TDMR read head **700**, four spin currents flow. A first spin current travels from the first SOT layer **510***a* down through the first non-magnetic layer **506***a* to the first free layer **418***a*; enabling an electrical signal generation when the first free layer **418***a* rotates magnetization directions when reading data. A second spin current travels from the second SOT layer **510***b* down through the second non-magnetic layer **506***b* to the second free layer **418***b* enabling an electrical signal generation when the second free layer **418***b* switches magnetization directions when reading data.
(85) A third spin current travels from the third SOT layer **610***a* down through the first non-magnetic layer **506***a* to the third free layer **718***a*, enabling an electrical signal generation when the third free layer **718***a* switches magnetization directions when reading data. A fourth spin current travels from the fourth SOT layer **610***b* down through the second non-magnetic layer **506***b* to the fourth free layer **718***b*, allowing the fourth spin current to flow to the fourth free layer **718***b* enabling an electrical signal generation when the fourth free layer **718***b* switches magnetization directions when reading data. The opposite magnetization direction of the second free layer **418***b* and the fourth free layer **718***b*, and of the first free layer **418***a* and the third free layer **718***a* enable generated signal adding together within each sensor **740***a* and **740***b*.
(86) FIGS. **8**A-**8**B illustrate a TDMR read head **800** of a magnetic recording head, according to another embodiment. FIG. **8**A illustrates a cross-sectional view of the TDMR read head **800** and FIG. **8**B illustrates a MFS view of the TDMR read head **800**. The TDMR read head **800** is similar

to the TDMR read head **700** of FIGS. **7**A-**7**B; however, the TDMR read head **800** comprises a shared common middle shield **803** rather than the first upper shield **404**, the second lower shield **502**, and the insulating layer **542**.
(87) As such, in the TDMR read head **800**, the middle shield **803** is disposed in contact with the first cap layer **522***a*, the third tunnel barrier layer **720***b*, the first SB side shield **524***a*, and the second SB side shield **526***a*. Like the TDMR read head **700**, from the first lower shield **402** to the second upper shield **504**, the first, second, third, and fourth free layers **418***a*, **418***b*, **718***a*, **718***b* have alternating magnetization directions. Each SOT layer **510***a*, **510***b*, **610***a*, **610***b* is spaced the distance **444** from the first free layer **418***a*, the second free layer **418***b*, the third free layer **718***a*, and the fourth free layer **718***b*, respectively.
(88) A first negative voltage lead (V.sub.1−) is connected to the first lower shield **402**, and a second positive voltage lead (V.sub.2+) is connected to the second upper shield **504**. A first positive voltage lead (V.sub.1+) and a second negative voltage lead (V.sub.2−) are each connected to the same middle shield **803**.
(89) A first positive current lead (I.sub.1+) and a first negative current lead (I.sub.1−) are each connected to the first SOT layer **510***a*. A first electrical current (I.sub.1) travels through the first SOT layer **510***a* into the page of FIG. **8**A, or the −z-direction. A second positive current lead (I.sub.2+) and a second negative current lead (I.sub.2−) are each connected to the second SOT layer **510***b*. A second electrical current (I.sub.2) travels through the second SOT layer **510***b* into the page of FIG. **8**A, or the −z-direction. A third positive current lead (I.sub.3+) and a third negative current lead (I.sub.3−) are each connected to the third SOT layer **610***a*. A third electrical c current (I.sub.3) travels through the third SOT layer **610***a* out of the page of FIG. **8**A, or in the z-direction. A fourth positive current lead (I.sub.4+) and a fourth negative current lead (I.sub.4−) are each connected to the fourth SOT layer **610***b*. A fourth electrical current (I.sub.4) travels through the fourth SOT layer **610***b* out of the page of FIG. **8**A, or in the z-direction.
(90) Thus, from the first lower shield **402** to the second upper shield **504**, the third SOT layer **610***a* has an electrical current traveling in the z-direction, the first SOT layer **510***a* has electrical current traveling in the −z-direction, the fourth SOT layer **610***b* has an electrical current traveling in the z-direction, and the second SOT layer **510***b* has an electrical current traveling in the −z-direction. As such, the first, second, third, and fourth SOT layers **510***a*, **510***b*, **610***a*, **610***b* have alternating electrical current directions.
(91) Moreover, the first positive current lead (I.sub.1+) is aligned with, or disposed adjacent to, to the third negative current lead (I.sub.3−), and the second positive current lead (I.sub.2+) is aligned with, or disposed adjacent to, to the fourth negative current lead (I.sub.4−). As such, the first negative current lead (I.sub.1−) is aligned with, or disposed adjacent to, to the third positive current lead (I.sub.3+), and the second negative current lead (I.sub.2−) is aligned with, or disposed adjacent to, to the fourth positive current lead (I.sub.4+). Thus, the positive and negative current leads are alternating.
(92) During operation, when currents as described above are applied to the TDMR read head **800**, four spin currents flow in the same manner as described above with respect to the TDMR read head **700** of FIGS. **7**A-**7**B.
(93) By including at least one SOT layer in a read sensor recessed from the MFS, the read heads discussed above are able to achieve a higher spin current injection/polarization, such that by making effective polarization significantly larger than 1. Furthermore, due to the trapezoidal shape of the non-magnetic transport layer, the spin current is more concentrated and the signal output is increased at the read sensor disposed at the MFS.
(94) In one embodiment, a read head comprises a first shield; a second shield, a first non-magnetic layer disposed between the first shield and the second shield, a first sensor disposed between the first non-magnetic layer and the second shield at a media facing surface (MFS), the first sensor comprising a first free layer, a first spin generator recessed from the MFS, the first spin generator

being spaced from the first sensor, wherein the first spin generator comprises a first spin orbit torque (SOT) layer, a third shield disposed over the second shield, a second non-magnetic layer disposed between the second shield and the third shield, a second sensor disposed between the second non-magnetic layer and the third shield at the MFS, the second sensor comprising a second free layer, and a second spin generator recessed from the MFS, the second spin generator being spaced from the second sensor, wherein the second spin generator comprises a second SOT layer.
(95) The read head further comprises a first non-magnetic notch disposed between the first non-magnetic layer and the first shield, and a second non-magnetic notch disposed between the second non-magnetic layer and the second shield. The read head further comprises a fourth shield disposed between the second shield and the second non-magnetic layer, and an insulating layer disposed between the second shield and the fourth shield. The read head further comprises a first negative voltage lead connected to the first shield, a first positive voltage lead connected to the second shield, a second negative voltage lead connected to the fourth shield, and a second positive voltage lead connected to the third shield. The read head further comprises a first positive current lead connected to the first SOT layer, a first negative current lead connected to the first SOT layer, a second positive current lead connected to the second SOT layer, and a second negative current lead connected to the second SOT layer. The read head further comprises a first negative current lead connected to the first shield, a first positive current lead connected to the second shield, a second negative current lead connected to the fourth shield, and a second positive current lead connected to the third shield. The read head further comprises a first negative current lead connected to the first non-magnetic notch, a first positive current lead connected to the second shield, a second negative current lead connected to the second non-magnetic notch, and a second positive current lead connected to the third shield.
(96) The read head further comprises a first soft bias side shield disposed adjacent to the first sensor at the MFS, a second soft bias side shield disposed adjacent to the first sensor at the MFS, wherein the first sensor is disposed between the first soft bias side shield and the second soft bias side shield, a third soft bias side shield disposed adjacent to the second sensor at the MFS, and a fourth soft bias side shield disposed adjacent to the second sensor at the MFS, wherein the second sensor is disposed between the third soft bias side shield and the fourth soft bias side shield. The first spin generator is disposed between the first non-magnetic layer and the second shield, and wherein the second spin generator disposed is between the second non-magnetic layer and the third shield. A magnetic recording head comprises the read head. A magnetic recording device comprises the magnetic recording head.
(97) In another embodiment, a magnetic recording head comprises a read head, the read head comprising: a first shield, the first shield comprising a first shield notch, a second shield disposed over the first shield, a first non-magnetic layer disposed between the first shield notch and the second shield, a first sensor disposed between the first non-magnetic layer and the second shield at a media facing surface (MFS), the first sensor comprising a first free layer, a first spin generator disposed between the first shield and the second shield recessed from the MFS, the first spin generator being spaced from the first sensor, wherein the first spin generator comprises: a first spin orbit torque (SOT) layer disposed between the first shield and the first non-magnetic layer, and a second SOT layer disposed between the first non-magnetic layer and the second shield, a third shield disposed over the second shield, the third shield comprising a second shield notch, a fourth shield disposed over the third shield, a second non-magnetic layer disposed between the second shield notch and the fourth shield, a second sensor disposed between the second non-magnetic layer and the fourth shield at the MFS, the second sensor comprising a second free layer, and a second spin generator disposed between the third shield and the fourth shield recessed from the MFS, the second spin generator being spaced from the second sensor, wherein the second spin generator comprises: a third SOT layer disposed between the third shield and the second non-magnetic layer, and a fourth SOT layer disposed between the second non-magnetic layer and the

fourth shield.
(98) The first SOT layer is disposed adjacent to the first shield notch, the second SOT layer is disposed adjacent to the first free layer, the third SOT layer is disposed adjacent to the second shield notch, and the fourth SOT layer is disposed adjacent to the second free layer. The read head further comprises a first negative voltage lead connected to the first shield, a first positive voltage lead connected to the second shield, a second negative voltage lead connected to the third shield, and a second positive voltage lead connected to the fourth shield. The read head further comprises a first negative current lead connected to the first SOT layer, a first positive current lead connected to the first SOT layer, a second positive current lead connected to the second SOT layer, a second negative current lead connected to the second SOT layer, a third negative current lead connected to the third SOT layer, a third positive current lead connected to the third SOT layer, a fourth positive current lead connected to the fourth SOT layer, and a fourth negative current lead connected to the fourth SOT layer.
(99) The read head further comprises a first non-magnetic notch disposed between the first shield notch and the first non-magnetic layer, a second non-magnetic notch disposed between the second shield notch and the third shield, a first soft bias (SB) side shield disposed adjacent to the first free layer, a second SB side shield disposed adjacent to the first free layer, a third SB side shield disposed adjacent to the second free layer, and a fourth SB side shield disposed adjacent to the second free layer. The read head further comprises a first negative current lead connected to the first shield, a first positive current lead connected to the second shield, a second negative current lead connected to the fourth shield, and a second positive current lead connected to the third shield. The read head further comprises a first negative current lead connected to the first non-magnetic notch, a first positive current lead connected to the second shield, a second negative current lead connected to the second non-magnetic notch, and a second positive current lead connected to the third shield. A magnetic recording device comprises the magnetic recording head.
(100) In yet another embodiment, a magnetic recording head comprises a read head, the read head comprising: a first shield, a second shield, a first non-magnetic layer disposed between the first shield and the second shield, a first sensor disposed between the first shield and the second shield at a media facing surface (MFS), the first sensor comprising a first free layer and a second free layer, a first spin generator disposed between the first shield and the second shield recessed from the MFS, the first spin generator being spaced from the sensor, wherein the first spin generator comprises: a first spin orbit torque (SOT) layer disposed between the first shield and the first non-magnetic layer, and a second SOT layer disposed between the first non-magnetic layer and the second shield, a third shield disposed over the second shield, a second non-magnetic layer disposed between the second shield and the third shield, a second sensor disposed between the second shield and the third shield at the MFS, the second sensor comprising a third free layer and a fourth free layer, and a second spin generator disposed between the second shield and the third shield recessed from the MFS, the second spin generator being spaced from the second sensor, wherein the second spin generator comprises: a third SOT layer disposed between the second shield and the second non-magnetic layer, and a fourth SOT layer disposed between the second non-magnetic layer and the third shield.
(101) The read head further comprises a fourth shield disposed between the second shield and the second non-magnetic layer, and an insulating layer disposed between the second shield and the fourth shield. The read head further comprises a first negative voltage lead connected to the first shield, a first positive voltage lead connected to the second shield, a second negative voltage lead connected to the third shield, and a second positive voltage lead connected to the fourth shield. The read head further comprises a first negative current lead connected to the first SOT layer, a first positive current lead connected to the first SOT layer, a second positive current lead connected to the second SOT layer, a second negative current lead connected to the second SOT layer, a third negative current lead connected to the third SOT layer, a third positive current lead connected to

the third SOT layer, a fourth positive current lead connected to the fourth SOT layer, and a fourth negative current lead connected to the fourth SOT layer.

(102) The read head further comprises a first negative voltage lead connected to the first shield, a first positive voltage lead connected to the second shield, a second negative voltage lead connected to the second shield, and a second positive voltage lead connected to the third shield. The read head further comprises a first negative current lead connected to the first shield, a first positive current lead connected to the second shield, a second negative current lead connected to the fourth shield, and a second positive current lead connected to the third shield. The first free layer is disposed between the first shield and the first non-magnetic layer, the second free layer is disposed between the first non-magnetic layer and the second shield, the third free layer is disposed between the second shield and the second non-magnetic layer, and the fourth free layer is disposed between the second non-magnetic layer and the third shield. The read head further comprises a first soft bias side shield disposed adjacent to the first free layer at the MFS, a second soft bias side shield disposed adjacent to the first free layer at the MFS, a third soft bias side shield disposed adjacent to the second free layer at the MFS, a fourth soft bias side shield disposed adjacent to the second free layer at the MFS, a fifth soft bias side shield disposed adjacent to the third free layer at the MFS, a sixth soft bias side shield disposed adjacent to the third free layer at the MFS, a seventh soft bias side shield disposed adjacent to the fourth free layer at the MFS, and an eighth soft bias side shield disposed adjacent to the fourth free layer at the MFS.

(103) During operation, the first free layer and the second free layer have anti-parallel magnetizations, the third free layer and the fourth free layer have anti-parallel magnetizations, and the first free layer and the third free layer have parallel magnetizations. The read head further comprises a fourth shield disposed between the second shield and the second non-magnetic layer, a first non-magnetic notch disposed between the first non-magnetic layer and the first shield, a second non-magnetic notch disposed between the second non-magnetic layer and the second shield, a first negative current lead connected to the first non-magnetic notch, a first positive current lead connected to the second shield, a second negative current lead connected to the second non-magnetic notch, and a second positive current lead connected to the third shield. A magnetic recording device comprises the magnetic recording head.

(104) While the foregoing is directed to embodiments of the present disclosure, other and further embodiments of the disclosure may be devised without departing from the basic scope thereof, and the scope thereof is determined by the claims that follow.

Claims

1. A read head, comprising: a first shield; a second shield; a first non-magnetic layer disposed between the first shield and the second shield; a first sensor disposed between the first non-magnetic layer and the second shield at a media facing surface (MFS), the first sensor comprising a first free layer; a first spin generator recessed from the MFS, the first spin generator being spaced from the first sensor wherein the first spin generator comprises a first seed layer, a first non-magnetic spin orbit torque (SOT) layer, and a first cap layer; a third shield disposed over the second shield; a second non-magnetic layer disposed between the second shield and the third shield; a second sensor disposed between the second non-magnetic layer and the third shield at the MFS, the second sensor comprising a second free layer; and a second spin generator recessed from the MFS, the second spin generator being spaced from the second sensor, wherein the second spin generator comprises a second seed layer, a second non-magnetic SOT layer, and a second cap layer.
2. The read head of claim 1, further comprising: a first magnetic notch disposed between the first non-magnetic layer and the first shield; and a second magnetic notch disposed between the second non-magnetic layer and the second shield.
3. The read head of claim 1, further comprising: a fourth shield disposed between the second shield

and the second non-magnetic layer; and an insulating layer disposed between the second shield and the fourth shield.

4. The read head of claim 3, further comprising: a first negative voltage lead connected to the first shield; a first positive voltage lead connected to the second shield; a second negative voltage lead connected to the fourth shield; and a second positive voltage lead connected to the third shield.

5. The read head of claim 3, further comprising: a first positive current lead connected to the first non-magnetic SOT layer; a first negative current lead connected to the first non-magnetic SOT layer; a second positive current lead connected to the second non-magnetic SOT layer; and a second negative current lead connected to the second non-magnetic SOT layer.

6. The read head of claim 1, further comprising: a first soft bias side shield disposed adjacent to the first sensor at the MFS; a second soft bias side shield disposed adjacent to the first sensor at the MFS, wherein the first sensor is disposed between the first soft bias side shield and the second soft bias side shield; a third soft bias side shield disposed adjacent to the second sensor at the MFS; and a fourth soft bias side shield disposed adjacent to the second sensor at the MFS, wherein the second sensor is disposed between the third soft bias side shield and the fourth soft bias side shield.

7. The read head of claim 1, wherein the first spin generator is disposed between the first non-magnetic layer and the second shield, and wherein the second spin generator disposed is between the second non-magnetic layer and the third shield.

8. A magnetic recording head comprising the read head of claim 1.

9. A magnetic recording device comprising the magnetic recording head of claim 8.

10. A magnetic recording head, comprising: a read head, the read head comprising: a first shield, the first shield comprising a first shield notch; a second shield disposed over the first shield; a first non-magnetic layer disposed between the first shield notch and the second shield; a first sensor disposed between the first non-magnetic layer and the second shield at a media facing surface (MFS), the first sensor comprising a first free layer; a first spin generator disposed between the first shield and the second shield recessed from the MFS, the first spin generator being spaced from the first sensor, wherein the first spin generator comprises: a first seed layer; a first non-magnetic spin orbit torque (SOT) layer disposed between the first shield and the first non-magnetic layer; a second non-magnetic SOT layer disposed between the first non-magnetic layer and the second shield; and a first cap layer; a third shield disposed over the second shield, the third shield comprising a second shield notch; a fourth shield disposed over the third shield; a second non-magnetic layer disposed between the second shield notch and the fourth shield; a second sensor disposed between the second non-magnetic layer and the fourth shield at the MFS, the second sensor comprising a second free layer; and a second spin generator disposed between the third shield and the fourth shield recessed from the MFS, the second spin generator being spaced from the second sensor, wherein the second spin generator comprises: a second seed layer; a third non-magnetic SOT layer disposed between the third shield and the second non-magnetic layer; a fourth non-magnetic SOT layer disposed between the second non-magnetic layer and the fourth shield; and a second cap layer.

11. The magnetic recording head of claim 10, wherein the read head further comprises: a first magnetic notch disposed between the first shield notch and the first non-magnetic layer; a second magnetic notch disposed between the second shield notch and the third shield; a first soft bias (SB) side shield disposed adjacent to the first free layer; a second SB side shield disposed adjacent to the first free layer; a third SB side shield disposed adjacent to the second free layer; and a fourth SB side shield disposed adjacent to the second free layer.

12. The magnetic recording head of claim 11, wherein the read head further comprises: a first negative current lead connected to the first magnetic notch; a first positive current lead connected to the second shield; a second negative current lead connected to the second magnetic notch; and a second positive current lead connected to the fourth shield.

13. The magnetic recording head of claim 10, wherein: the first non-magnetic SOT layer is

disposed adjacent to the first shield notch; the second non-magnetic SOT layer is disposed adjacent to the first free layer; the third non-magnetic SOT layer is disposed adjacent to the second shield notch; and the fourth non-magnetic SOT layer is disposed adjacent to the second free layer.
14. The magnetic recording head of claim 10, wherein the read head further comprises: a first negative voltage lead connected to the first shield; a first positive voltage lead connected to the second shield; a second negative voltage lead connected to the third shield; and a second positive voltage lead connected to the fourth shield.
15. The magnetic recording head of claim 10, wherein the read head further comprises: a first negative current lead connected to the first non-magnetic SOT layer; a first positive current lead connected to the first non-magnetic SOT layer; a second positive current lead connected to the second non-magnetic SOT layer; a second negative current lead connected to the second non-magnetic SOT layer; a third negative current lead connected to the third non-magnetic SOT layer; a third positive current lead connected to the third non-magnetic SOT layer; a fourth positive current lead connected to the fourth non-magnetic SOT layer; and a fourth negative current lead connected to the fourth non-magnetic SOT layer.
16. The magnetic recording head of claim 10, wherein the read head further comprises: a first negative current lead connected to the first shield; a first positive current lead connected to the second shield; a second negative current lead connected to the third shield; and a second positive current lead connected to the fourth shield.
17. A magnetic recording device comprising the magnetic recording head of claim 10.
18. A read head, comprising: a first shield; a second shield; a first non-magnetic layer disposed between the first shield and the second shield; a first sensor disposed between the first non-magnetic layer and the second shield at a media facing surface (MFS), the first sensor comprising a first free layer; a first spin generator recessed from the MFS, the first spin generator being spaced from the first sensor, wherein the first spin generator comprises a first spin orbit torque (SOT) layer; a third shield disposed over the second shield; a second non-magnetic layer disposed between the second shield and the third shield; a second sensor disposed between the second non-magnetic layer and the third shield at the MFS, the second sensor comprising a second free layer; a second spin generator recessed from the MFS, the second spin generator being spaced from the second sensor, wherein the second spin generator comprises a second SOT layer; a fourth shield disposed between the second shield and the second non-magnetic layer; an insulating layer disposed between the second shield and the fourth shield; a first negative current lead connected to the first shield; a first positive current lead connected to the second shield; a second negative current lead connected to the fourth shield; and a second positive current lead connected to the third shield.
19. A read head, comprising: a first shield; a second shield; a first non-magnetic layer disposed between the first shield and the second shield; a first sensor disposed between the first non-magnetic layer and the second shield at a media facing surface (MFS), the first sensor comprising a first free layer; a first spin generator recessed from the MFS, the first spin generator being spaced from the first sensor, wherein the first spin generator comprises a first spin orbit torque (SOT) layer; a third shield disposed over the second shield; a second non-magnetic layer disposed between the second shield and the third shield; a second sensor disposed between the second non-magnetic layer and the third shield at the MFS, the second sensor comprising a second free layer; a second spin generator recessed from the MFS, the second spin generator being spaced from the second sensor, wherein the second spin generator comprises a second SOT layer; a fourth shield disposed between the second shield and the second non-magnetic layer; a first negative current lead connected to a first magnetic notch; a first positive current lead connected to the second shield; a second negative current lead connected to a second magnetic notch; and a second positive current lead connected to the third shield.